

9

STATISTICS AND PROBABILITY

ANALYSING DATA

Statistical data are used by governments and businesses to make decisions about health and welfare, the population, production, tourism and the environment. Data collected from a census, through market research (such as online surveys, questionnaires, street polls) can lead to advertising campaigns for health issues and road safety, improved customer service and new products. However, before any action can be taken, the data must be carefully analysed in order to avoid misinterpretation and to check it for relevance and reliability.



Shutterstock.com/ilkercelik

Chapter outline

	Working mathematically				
9.01 The mean, median, mode and range	U	F		R	C
9.02 Histograms and stem-and-leaf plots	U	F		R	C
9.03 The shape of a distribution	U	F		R	C
9.04 Comparing data sets	U	F		R	C
9.05 Sampling and types of data	U	F		R	C
9.06 Bias and questionnaires*	U	F	PS	R	C

***PATH**

Wordbank

bias Something unwanted that causes a sample to not truly represent the population

cluster Values in a data set that are close or 'bunched' together

frequency histogram A special column graph that shows the frequencies of numerical data

outlier An extreme value that is very different from the other values in a set of data

population All of the items being studied, the entire group

random sample A sample in which every item in the population has an equal chance of being selected

skewed distribution A distribution in which most of the values are clustered at one end, creating a 'tail' at the other end

symmetrical distribution A distribution in which all values are distributed equally on both sides of the centre, its shape having line symmetry

Quiz
Wordbank 9

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (8):

SkillCheck Statistics

9.01 The mode, median and mean • Analysing dot plots • Statistics from a frequency table • Cumulative frequency and the median

9.02 Frequency histograms and polygons • Back-to-back stem-and-leaf plots

9.03 The shape of a frequency distribution

Twig videos (6):

9.01 Cumulative frequency: You're fired?

9.02 Histograms: snapshot

9.05 Counting crowds • Can fish oil make you smarter? • Can you trust your IQ?

9.06 The wrong guy won

PhET interactive (1):

9.01 Centre and variability

Quizzes (5):

- Wordbank 9
- SkillCheck 9
- Mental skills 9
- Language of maths 9
- Test yourself 9

Skillsheet (1):

9.01 Statistical measures

Worksheets (9):

9.01 Mean, median and mode • Collecting and describing data • The mean and *fx* tables

9.02 Statistics review • Analysing data • Stem-and-leaf plots

9.04 Comparing word lengths • Investigating young drivers

Mind map Analysing data

Puzzles (5):

9.01 Mode, median and mean

9.02 Highway elephants

9.03 Statistical match-up

9.05 Data match-up

Language of maths Statistics crossword

Technology (3):

9.01 Statistical measures calculator

9.04 Population polygons • World climate

Spreadsheets (2):

9.01 Statistical measures calculator

9.04 World climate

Nelson MindTap

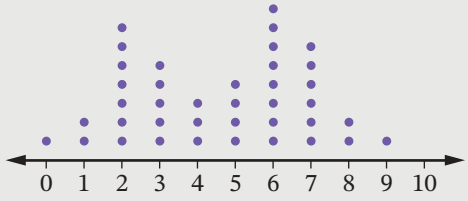
To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- calculate and interpret the mean, median, mode and range for a set of data
- investigate the effect of an outlier on the mean and median
- construct histograms and stem-and-leaf plots
- describe the shape of a statistical distribution by identifying skewness, symmetry and modality
- compare 2 or more sets of data by analysing their graphs and statistics
- (PATH) investigate sampling compared to a census
- identify categorical and numerical data, sort categorical data into ordinal and nominal data, and sort numerical data into discrete and continuous data
- (PATH) evaluate statistics found in the media and detect bias in reports of surveys

SkillCheck ANSWERS ON P. 635

1 This dot plot shows the number of apps purchased by students last month.



Number of apps purchased

a How many students were surveyed?

b What was the most frequent number of apps purchased?

c What percentage (correct to one decimal place) of students purchased over 5 apps?

2 This stem-and-leaf plot shows the ages of people at a café during 8:00–8:30 a.m. on a Saturday morning.

Stem	Leaf
1	4 6 7 9 9
2	0 4 5 7 8 8
3	1 5 5 6 6 6 7 8 9
4	2 3 3 5 7
5	0 0 1 4 5 6 8 8
6	3 5 6 7 7

a How many people were in the café?

b How many of them were aged in their 20s?

c What was the most common age?

d What was the youngest age?

e What percentage (correct to one decimal place) of people were 45 or older?

3 The length of a queue at a supermarket checkout was recorded every 5 minutes for 3 hours.

4	3	6	5	1	4	7	5	4	1	4	4
4	3	6	3	1	3	4	2	4	5	1	3
5	4	2	3	5	2	5	2	6	5	1	3

a Arrange the data in a frequency table.

b Draw a frequency histogram and a frequency polygon.

c What was the most common number of people in the queue?

d How many times was the length of the queue longer than 3 people?

Quiz
SkillCheck 9

Video
Statistics

9.01

9.01 The mean, median, mode and range

Video
The mode, median and mean

Worksheets
Mean, median and mode

Collecting and describing data

Skillsheet
Statistical measures

Puzzle
Mode, median and mean

Interactive
Centre and variability

Technology
Statistical measures calculator

Spreadsheet
Statistical measures calculator

The **mean**, **median** and **mode** are called **measures of centre**, **measures of central tendency**, **measures of location** or **averages**, because they indicate a central value of a set of **data**.

The **range** is called a measure of spread because it indicates how widely the values are spread in a set of data.

Measures of centre and spread

Mean, $\bar{x} = \dfrac{\text{sum of data values}}{\text{number of data values}}$

$= \dfrac{\sum x}{n}$

where Σx means 'the sum of the values' and n is the total number of values.

Σ is the Greek letter **sigma**, and stands for 'the sum of'.

The **mode** is the value (or values) that occurs most often.

When the data values are arranged in order, the **median** is:

- the middle value, for an odd number of values
- the average of the 2 middle values, for an even number of values.

Range = highest value – lowest value

An **outlier** (pronounced 'out-ly-er') is an extremely high or low value that is separated from the other values in the data set.

Comparing the mean, median and mode

The **mean** is usually the best measure of centre because it takes into account every data value.

If there are outliers in the set of data, then the mean may be affected by these extreme values and will not accurately represent all of the values. In this case, the **median** is the best measure because it is not affected by outliers.

The **mode** is useful when the most common value is important, or when the data is categorical (such as make of car or hair colour). With categorical data, it is not possible to have a mean or median.

Example 1

The maximum daily temperatures (in °C) in Sydney for a fortnight in March were:

34	22	32	26	23	24	24
26	31	26	27	28	28	27

Find:

a the mean

b the median

c the mode

d the range

Solution

a $\text{Mean} = \frac{34+22+32+\cdots+28+27}{14}$

$= \frac{378}{14}$

$= 27$

b Arranging the values from lowest to highest: the middle values

22 23 24 24 26 26 26 27 27 28 28 31 32 34

There is an even number of values (14), so we have 2 middle values, the 7th and 8th values 26 and 27.

$\text{Median} = \frac{26+27}{2}$

$= 26.5$

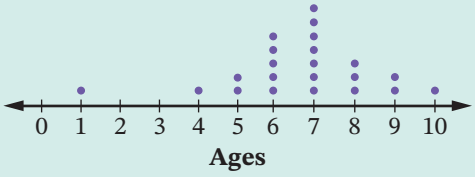
c Mode = 26 The most common value, occurring 3 times

d Range = 34 – 22 Highest value – lowest value

$= 12$

Example 2

This **dot plot** shows the ages of the children at Soshanna's birthday party.



Find:

a the mean (correct to one decimal place)

b the median

c the mode

d the range

e the outlier

Solution

a $\text{Mean} = \frac{1+4+5+5+\cdots+10}{22}$

$= 6.636363...$

≈ 6.6

b There are 22 values, so the median is the average of the 11th and 12th values. Counting dots from the left and upwards, these are 7 and 7.

$\text{Median} = \frac{7+7}{2} = 7$

c Mode = 7 The value with the most dots

d Range = 10 – 1 = 9

e Outlier = 1 This value is much different from the rest

Video
Analysing dot plots

The mean from a calculator's statistics mode

Scientific and graphics calculators have a statistics mode (SD or STAT). Follow the instructions in the table below to calculate the mean of the data from Example **1** using your calculator's statistics mode.

Operation	Casio	Sharp
Start statistics mode.	MODE STAT 1-VAR	MODE STAT =
Clear the statistical memory.	SHIFT 1 Edit, Del-A	2nd F DEL
Enter data.	SHIFT 1 Data to get table 34 = 22 = , etc. to enter in column AC to leave table	34 M+ 22 M+, etc.
Calculate the mean. ($\bar{x}$ = 27:0714)	SHIFT 1 Var $\bar{x}$ =	RCL $\bar{x}$
Check the number of values. (n = 14)	SHIFT 1 Var n =	RCL n
Return to normal (COMP) mode.	MODE COMP	MODE 0

The mean from a frequency table

Worksheet
The mean and fx tables

For a large set of data, we can use a **frequency distribution table** to organise the data. By adding an fx column, the mean can be easily found.

Score (x)	Frequency (f)
3	2
4	5
5	10
6	36
7	25
8	13
9	4
10	3

Video
Statistics from a frequency table

Example 3

The scores of Year 9 students giving a speech in English (out of 10) are shown in the frequency table above.

a How many students were there?

b Use an fx column to calculate the mean score.

c Find the mode.

d What is the range?

Solution

a The total sum of the Frequency column is 98.

There were 98 students.

b

Mark (x)	Frequency (f)	fx
3	2	6
4	5	20
5	10	50
6	36	216
7	25	175
8	13	104
9	4	36
10	3	30
	$\sum f = 98$	$\sum fx = 637$

fx means $f \times x$, which multiplies each value by its frequency.

$\sum f$ is the sum of the frequency column (that is, the total number of values).

$\sum fx$ is the sum of the fx column (that is, the sum of all values).

$$\bar{x} = \frac{\sum fx}{\sum f}$$

$$\text{Mean} = \frac{637}{98} = 6.5$$

Sum of all values divided by the number of values

c Mode = 6

Value with the highest frequency

d Range = 10 − 3 = 7

Mean, mode and range for data in a frequency table

- **Mean**, $\bar{x} = \frac{\text{sum of } fx}{\text{sum of } f} = \frac{\sum fx}{\sum f}$
- The **mode** is the value(s) with the highest frequency, f
- The **range** is the difference between the last and first values in the score (x) column

This table shows calculator instructions for finding the mean of data presented in a frequency table. It uses the table of data from Example **3**.

Operation	Casio	Sharp
Start statistics mode.	MODE STAT 1-VAR SHIFT MODE scroll down to STAT Frequency? ON	MODE STAT =
Clear the statistical memory.	SHIFT 1 Edit, Del-A	2nd F DEL
Enter data.	SHIFT 1 Data to get table 3 = 4 =, etc. to enter in x column 2 = 5 =, etc. to enter in FREQ column AC to leave table	3 2nd F STO 2 M+ 4 2nd F STO 5 M+, etc.
Calculate the mean. ($\bar{x}$ = 6.5)	SHIFT 1 Var $\bar{x}$ =	RCL $\bar{x}$
Check the number of values. (n = 98)	SHIFT 1 Var n =	RCL n
Return to normal (COMP) mode.	MODE COMP	MODE 0

Cumulative frequency and the median

Videos
Cumulative frequency: You're fired?

Cumulative frequency and the median

To find the median of data presented in a frequency table, add a **cumulative frequency** column, which is a progressive total of the frequency column.

Example 4

Find the median of the Year 9 student speech scores from Example **3**.

Solution

The cumulative frequency column is a running total of the frequencies.

$2 + 5 = 7$

$7 + 10 = 17$, etc.

There are 98 scores, so the 2 middle values are the 49th and 50th scores. Using the cumulative frequency column, the 17th score is a 5 and the 53rd score is 6, so the 49th and 50th scores must both be 6s.

$\text{Median} = \frac{6+6}{2} = 6$

Mark (x)	Frequency (f)	Cumulative frequency (cf)
3	2	2
4	5	7
5	10	17
6	36	53
7	25	78
8	13	91
9	4	95
10	3	98
	$\Sigma f = 98$	

EXERCISE 9.01 ANSWERS ON P. 635

The mean, median, mode and range **U F R C**

Round all means to one decimal place where necessary, unless otherwise specified.

EXAMPLE 1

1 For each set of data, find:

i the mean **ii** the median **iii** the mode **iv** the range

a	23	20	12	15	28	19	15	14	18		
b	5	3	8	7	1	6	7	7	5	15	12
c	32	45	12	45	38	32					
d	8.8	8.2	8.8	8.3	8.5	8.9	9.2	8.8			
e	−5	0	−2	4	6	−2	5	−1	−2	0	

2 The maximum and minimum daily temperatures (in °C) over a week at Point Perpendicular on the NSW south coast are listed.

Maximum	23	24	22	23	22	23	23
Minimum	17	17	15	16	15	15	14

a Calculate the mean daily maximum and mean daily minimum temperature.

b What was the difference between the 2 means?

c Find the mode of daily maximum and minimum temperatures.

d What was the difference between the 2 modes?

Foundation Standard Complex

9780170479585

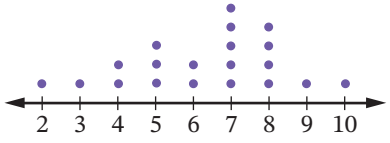
3 The monthly rainfall (in mm) in Cairns is shown in the table.

Month	J	F	M	A	M	J	J	A	S	O	N	D
Rainfall	413	435	442	191	94	49	28	27	36	38	90	175

a What is the range? **b** Find the mean (correct to the nearest mm).

c Find the median.

4 A number of homes were surveyed on how many digital devices they had, with the results shown in the dot plot.



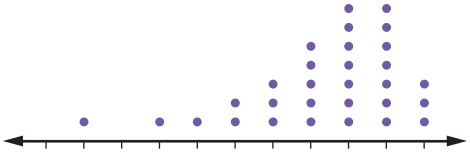
Number of devices

Find:

a the mean **b** the median

c the mode **d** the range

5 This dot plot shows the results of a maths quick quiz (out of 10). R C



Quiz results

a Are there any outliers? (Give reasons)

b What is the mode?

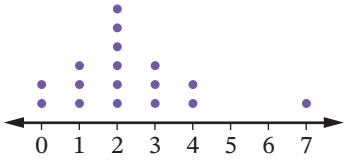
c Find the mean.

d Find the median.

e What percentage of results (correct to one decimal place) were above:

i the mean? **ii** the median?

6 This dot plot shows the number of goals scored by a soccer team during the season.



Number of goals

a Are there any outliers?

b Find the mean and median:

i without the outlier

ii with the outlier

c Which measure (mean or median) is affected by the outlier?

7 The salaries of 10 people working at a software company are:

$80 000	$105 000	$70 000	$93 000	$78 500
$180 000	$90 000	$72 000	$93 000	$101 000 R C

a Which salary is an outlier?

b Find the mean and median:

i without the outlier **ii** with the outlier

c Which measure, the mean or the median, represents the salaries better?

d Find the mode and range:

i without the outlier **ii** with the outlier

e Which measure, the mode or the range, is affected by the outlier?

EXAMPLE 2

9.01

Foundation Standard Complex

8 Use your calculator in the statistics mode to find the mean of each set of data.

a	5	7	8	3	6	8	5	4
	8	3	8	10	6	8	4	9
b	13.5	14.2	11.9	15.7	18.4	17.6	12.0	
c	56	72	84	57	49	65	59	
	79	80	76	57	66	68		

9 Cameron played in a golf tournament. His average score for the 4 rounds was 71. If the scores in his first 3 rounds were 72, 66 and 70, what was his score in the 4th round? R

10 Tasnuba scored a mean of 74% for 5 maths tests that she completed. R

a What was the total number of marks Tasnuba scored for the 5 tests?

b Tasnuba did a 6th test and her mean test mark increased to 77%.

What is the total that she scored for the 6 tests?

c What mark did she achieve in the last test?

11 Alice's average mark for 4 Science exams is 68%. Can Alice score enough marks in her 5th exam so that her average mark increases to 75%? Give reasons. R C

EXAMPLE 3

12 Copy and complete each frequency table, then calculate the mean, correct to 2 decimal places.

a

Score, x	Frequency, f	fx
2	5	
3	3	
4	7	
5	6	
6	4	
	Σf =	Σfx =

b

x	f	fx
22	3	
23	8	
24	7	
25	9	
26	5	
27	4	
	Σf =	Σfx =

13 Use your calculator in the statistics mode to find the mean of each frequency table.

a

Score, x	Frequency, f
32	5
33	8
34	12
35	9
36	7
37	4

b

x	f
10	12
11	18
12	27
13	37
14	23
15	15
16	9

EXAMPLE 4

9.01

14 For each frequency table, add a cumulative frequency column and find:

i the median **ii** the mode **iii** the range

a

x	f
2	2
3	5
4	11
5	7
6	10
7	9
8	6

b

x	f
17	3
18	8
19	20
20	15
21	11
22	7

15 The number of seedlings that have sprouted in each punnet at a nursery are shown:

10	5	7	7	9	8	8	11	7	8	5	8	4	6	4
12	11	9	4	6	6	10	7	6	9	10	9	7	8	8
8	7	4	7	7	11	10	9	8	9	9	8	8	8	9

a Complete a frequency table and find the median.

b What is the mode?

c Find the range.

d Calculate the mean.

Investigation

The effect of outliers

1 Find the mean, median, mode and range of each data set.

a 1 2 2 2 3 4 5 5 5

b 1 2 2 2 3 4 5 5 6

c 1 2 2 2 3 4 5 5 8

d 1 2 2 2 3 4 5 5 17

2 The 4 sets of data in question **1** are the same, apart from the last value in each set. If one value is changed, what is the effect on:

a the mean? **b** the median? **c** the mode? **d** the range?

3 Consider these 2 sets of data:

A	1	10	11	14	15	12	13	11		
B	3	4	2	6	3	14	6	3	5	4

a Find the mean, median and range of each set of data.

b Identify the outlier for each set.

c Find the mean, median and range of each set of data, excluding the outlier.

d What effect does the size of the outlier have on:

i the mean? **ii** the median? **iii** the range?

e Why is the median not affected by the size of the outlier?

Foundation Standard Complex

Technology

Most valuable player

A team of basketball players would like to compare the number of points each player has scored per game over 10 games.

1 Enter the following data into a spreadsheet.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	NAME	Game 1	Game 2	Game 3	Game 4	Game 5	Game 6	Game 7	Game 8	Game 9	Game 10	Total	Mean	Mode	Median	Range
2	Sirisha	5	8	2	10	6	8	4	4	6	10					
3	Colin	4	8	6	7	7	9	10	6	5	8					
4	Eloise	12	8	8	16	4	5	7	10	11	8					
5	Diana	4	3	5	2	6	8	4	3	3	6					
6	Tran	10	14	7	8	12	16	10	10	8	6					
7	Michelle	12	4	6	10	9	9	5	16	8	12					
8	Anish	4	7	6	8	7	5	3	8	9	2					
9	Jamar	14	12	8	7	10	9	5	12	15	8					
10	Total															
11	Mean															

2 In B10, enter **=sum(B2:B9)** to calculate the **total score** for Game 1. **Fill Right** to repeat this step for cells C10 to K10.

3 In L2, enter **=sum(B2:K2)** to calculate the **total score** for Sirisha for the 10 games. **Fill Down** to repeat this step for cells L3 to L9.

4 In B11, enter **=average(B2:B9)** to calculate the **mean** score per player for Game 1. **Fill Right** from cell B11 across to K11.

5 In M2, enter **=average(B2:K2)** to calculate the **mean** score for Sirisha for the 10 games. **Fill Down** into cells M3 to M9.

6 In N2, enter **=mode(B2:K2)** to calculate the **mode** score for Sirisha. **Fill Down** to cell N9.

7 In column O, use the formula **=median(B2:K2)** and **Fill Down**.

8 There is no formula for the **range**, but we can create one using the formulas for the **maximum** and **minimum** values. In P2, enter **=max(B2:K2)–min(B2:K2)** for Sirisha's range, then **Fill Down**.

9 Now answer these questions, writing each answer in your spreadsheet in the given cell [ ].

- **a** Which player had the best mean score? [A13]
- **b** Which player was the most inconsistent? Why? [A14]
- **c** Which of the statistics is the most useful for describing the players' overall contribution to team score? [A15]
- **d** Who do you think was the MVP (Most Valuable Player)? Why? [A16]

Histograms and stem-and-leaf plots 9.02

Frequency histograms and polygons

A **frequency histogram** is a column graph of numerical data where the columns stand together without gaps between them.

A **frequency polygon** is a line graph that is constructed by joining the midpoints of the tops of the columns of a frequency histogram, starting and ending on the horizontal axis. It is called a 'polygon' because the graph and the horizontal axis together make a shape with straight sides.

Worksheets
Statistics review
Analysing data
Stem-and-leaf plots

Puzzle
Highway elephants

Videos
Frequency histograms and polygons
Histograms: snapshot

Example 5

45 students were surveyed on the number of hours of homework they did in the past week. The results were as follows:

3	7	5	8	8	7	7	4	8	7	9	8	8	5	7
4	2	5	3	8	6	6	7	9	6	5	4	9	8	4
2	3	6	7	8	2	8	8	4	5	6	8	7	2	5

a Arrange the data in a frequency table.

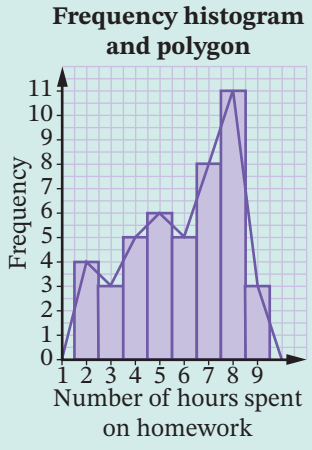
b Draw a frequency histogram and polygon for this data.

Solution

a

No. of hours, x	Tally	Frequency, f
2	IIII	4
3	III	3
4	𝍸	5
5	𝍸 I	6
6	𝍸	5
7	𝍸 III	8
8	𝍸 𝍸 I	11
9	III	3
	Total	45

b



Note that each column of the histogram is centred on each value marked on the horizontal axis, so there is a half-column gap at the start between the Frequency axis and the first column.

Stem-and-leaf plots

A **stem-and-leaf plot** lists the values of a data set, usually in order. It shows:

- any **clusters** where values are grouped or bunched together
- any **outliers**
- how the values are spread out

Example 6

This stem-and-leaf plot shows the pulse rate (in heartbeats per minute) of 30 people in a shopping centre.

The stem is formed by the 10s digit of the values.

Stem	Leaf
5	1 5 8
6	1 3 4 4 5 6 8 8 9
7	0 1 2 3 3 4 4 5 7 8 8 9
8	0 3 4 6 7
9	0

The leaves are formed by the units digits of the values.

The data shown by this row is 80, 83, 84, 86 and 87.

a Are there any outliers?

b Is there any clustering of data?

c Find the mean, correct to one decimal place.

d What is the median?

e What is the mode?

f What is the range?



iStock.com/Deklofenak

Solution

a There are no outliers as no values are 'separate' from the main body of data.

b The values are bunched in the 60s and 70s, so we say that there is clustering in the 60s and 70s.

c $\text{Mean} = \frac{51+55+58+\cdots+87+90}{30}$

$= \frac{2156}{30}$

$= 71.8666...$

≈ 71.9

d The 30 data values in the stem-and-leaf plot are in order, so the median is the average of the 15th and 16th values, 72 and 73 (counting from top to bottom, left to right).

$\text{Median} = \frac{72+73}{2} = 72.5$

e Modes = 64, 68, 73, 74, 78 Each occurs 2 times.

f The lowest value is 51 and the highest is 90.

Range = 90 – 51

= 39

Back-to-back stem-and-leaf plots

When 2 related sets of data are compared, we can use back-to-back stem-and-leaf plots.

Videos
Back-to-back stem-and-leaf plots

9.02

Example 7

Buzz batteries and *Eternal* batteries are to be compared by testing 15 batteries of each brand until they fail. The lengths of the battery lives are recorded to the nearest hour as shown.

Buzz	8	12	22	18	23	20	11	15	18	8	9	19	26	30	31
Eternal	14	16	24	32	35	36	37	28	31	30	24	26	19	35	32

a Arrange the data into a back-to-back stem-and-leaf plot to determine which battery is better.

b Describe any clusters in each distribution.

c Compare the median of each brand.

d Compare the mean of each brand.

e Which brand of battery is better? Justify your answer.

Solution

a

Buzz		Eternal
9 8 8	0	
9 8 8 5 2 1	1	4 6 9
6 3 2 0	2	4 4 6 8
1 0	3	0 1 2 2 5 5 6 7

b Buzz is clustered in the 10–20 hours' life, Eternal is clustered in the 30s hours of life.

c Median for Buzz = 18 — The 8th value, counting from top to bottom, right to left

Median for Eternal = 30 — The 8th value, counting from top to bottom, left to right

Eternal has the higher median of 30 hours.

d Mean for Buzz = 18

Mean for Eternal = 27.933...

Eternal has the higher mean of approximately 27.9 hours.

e From looking at the statistics and the stem-and-leaf plot, Eternal is better as its median and mean are higher.

EXERCISE 9.02 ANSWERS ON P. 636

Histograms and stem-and-leaf plots UFRC

EXAMPLE 5

1 The results of an Art project (out of 10) for 30 students are shown.

7	9	3	5	6	6	7	5	2	8
7	8	6	5	5	4	3	6	6	7
8	10	7	5	8	9	8	5	4	5

a Arrange the data into a frequency distribution table.

b Draw a frequency histogram and polygon.

c Find the mean, correct to one decimal place.

d Find the median.

e Find the mode.

f Find the range.

g Are there any outliers and is there any clustering? C

2 A sample of people were surveyed on the number of children in their families.

1	2	6	5	3	3	2	1	7	4	4	4	7	2
3	5	3	1	2	1	1	2	7	4	2	2	3	6
3	6	5	3	5	7	9	8	3	2	3	4		

a Arrange the data into a frequency table and construct a frequency histogram and polygon.

b Find the mode and the range.

c Are there any outliers and is there any clustering?

d Find the median.

e Find the mean, correct to one decimal place.

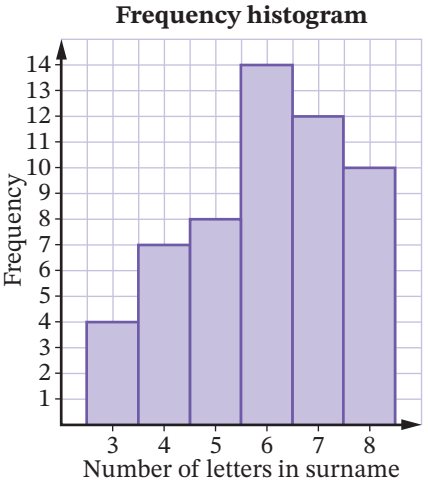
3 The numbers of letters in the surnames of 55 students are shown in this frequency histogram.

a What is the mode? How is this shown on the frequency histogram?

b Find the mean, correct to 2 decimal places.

c Find the median.

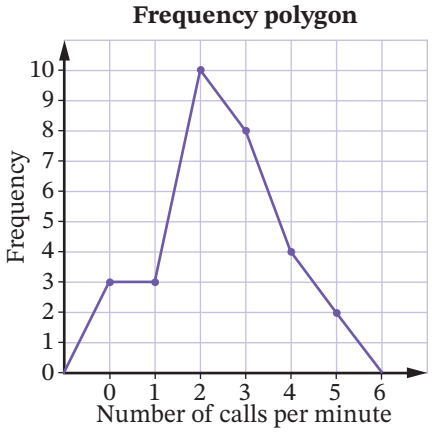
d Find the range.



Foundation Standard Complex

9780170479585

4 This frequency polygon shows the number of calls made per minute to a customer support line over a 30-minute period. R C



a How many calls were made altogether?

b How many times were there 3 or more calls received per minute?

c What is the mode? How is this shown on the frequency polygon?

d Find the range.

e Find the median.

f Find the mean, correct to 2 decimal places.

g Is there any clustering of data?

5 This unordered stem-and-leaf plot shows the marks of a class of students in a Science test.

Stem	Leaf
4	0 1 5
5	3 1 5 6 8
6	7 7 8 6 4 3 5 7
7	9 1 7 4 2
8	8 2 7
9	2 0 4

EXAMPLE 6

a Construct an ordered stem-and-leaf plot.

b Find the range.

c Are there any outliers or clusters?

d How many students are there in the class?

e Find the mean, correct to 2 decimal places.

f Find the median.

6 This stem-and-leaf plot shows the points scored by a basketball team in different matches.

Stem	Leaf
3	0 7
4	1 1 3 5
5	4 6 6 8 9
6	3 5 8 8 8 9
7	2 4 5 5
8	0 2

a Find the range.

b What is the mode?

c Find the mean, correct to one decimal place.

d Find the median.

7 When 30 students ran 100 m, their times (in seconds) were recorded.

13.5	15.0	12.1	12.5	17.6	13.7	15.7	18.7	13.1	11.8
12.6	15.7	14.4	15.4	14.9	15.0	18.3	16.3	15.4	15.1
11.9	16.1	14.3	14.5	16.3	14.6	14.9	15.0	16.1	13.5

Stem	Leaf
11	8 9
12	
13	
14	
15	
16	
17	
18	

a Copy and complete this stem-and-leaf plot.

b Are there any:

i outliers? **ii** clusters?

c How many students ran the 100 m in less than 13.7 seconds?

d What percentage of students took more than 14 seconds to run the 100 m?

e What percentage (to one decimal place) of students ran times that were below:

i the mean **ii** the median?

Foundation Standard Complex

8 This ordered stem-and-leaf plot shows the marks obtained by students in a History exam.

Stem	Leaf
2	□
3	0 2 3 4
4	1 5 7 8
5	2 4 □ 6 9 9
6	0 2 3 3 6 8 8 7
7	2 5 6 7 7
8	2 5

a One of the marks in the 50s is missing. What could this missing mark be?

b The entry for the lowest mark is missing. If the range of the marks is 63, find the lowest mark. R

c Find the median.

EXAMPLE 7

9 This back-to-back stem-and-leaf plot shows the number of goals scored per match by 2 netball teams during a season.

Rockets		Blues
7 5 4 3	1	7
8 7 6 5 4 2	2	0 5 6 8
7 6 4 3	3	4 5 7 7 8 9
5 4 0	4	2 3 7 8
2	5	1 3 7

a How many times did each team score more than 30 goals?

b Find the highest number of goals scored by each team.

c Which team had the greater range?

d Which team performed better? Give reasons. R C

e Which team had the greater median?

10 The number of hamburgers made by Rashmi and Anthony at lunchtime over 14 days was: R

Rashmi	37	23	33	35	17	42	37	54	45	42	37	27	36	41
Anthony	35	28	29	36	43	37	55	42	47	51	36	34	35	39

a Construct an ordered back-to-back stem-and-leaf plot.

b Who made more hamburgers?

c On how many days did Rashmi make more hamburgers than Anthony?

d Are there any outliers in the figures for either Rashmi or Anthony?

e Is there any clustering of figures for either person?

f Which person had the higher mean?

11 A sample of commuters were surveyed on how far (in kilometres) they live from the closest railway station. R C

1.5	4.8	6.5	1.1	3.3	3.4	3.5	4.5	4.5	12.5	3.1	3.6
3.8	4.0	4.2	2.3	2.5	1.9	4.7	5.5	2.7	3.2	5.2	

a Arrange the data in a stemplot.

b Are there any outliers? Give reasons.

c Are there any clusters?

d Find the mean, correct to 2 decimal places.

e Find the median.

f Which is the better measure of centre, the mean or median? Give reasons.

Foundation Standard Complex

9780170479585

12 The results for a writing task for 20 girls and 20 boys are shown.

Girls:	18	48	56	59	67	73	76	94	76	78
	82	39	45	59	60	85	80	74	51	36
Boys:	50	88	76	72	10	28	40	95	74	63
	67	55	32	29	16	75	26	82	57	50

a Construct a back-to-back stem-and-leaf plot for the data.

b What was the highest mark? Was it scored by a girl or boy?

c What is the median for:

i girls **ii** boys?

d What percentage of boys scored more than 60?

e What percentage of girls scored more than 60?

f Find the mean mark for:

i girls **ii** boys.

g Is there a there a significant difference between the median mark and the mean mark for:

i girls? **ii** boys?

Technology

Histograms

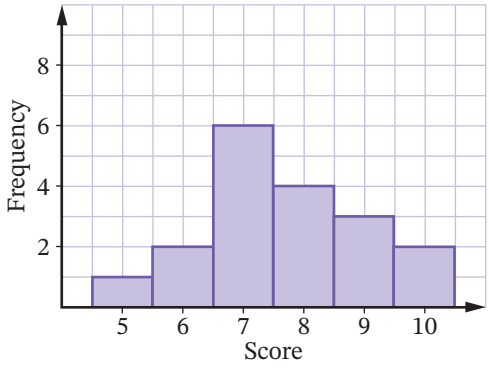
In this activity, we will use statistical, spreadsheet or dynamic geometry software to construct a histogram of players' scores (out of 10) in a trivia quiz.

Score	Frequency
5	1
6	2
7	6
8	4
9	3
10	2

1 Enter the data shown.

2 Your histogram should also have a heading and the axes labelled as shown below.

Players Scores (out of 10) in a Trivia Quiz



3 Remember, the data represents one-variable statistics. In the menu, select **1VAR** (or similar) to show calculations such as mean, median and mode.

Foundation Standard Complex

9.03 The shape of a distribution

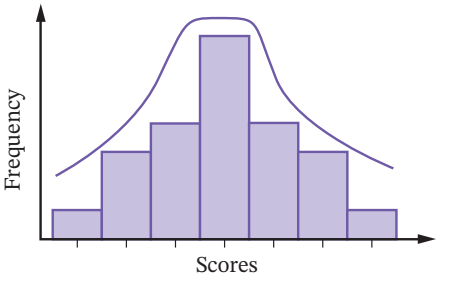
Puzzle
Statistical match-up

A **statistical distribution** is the way the values of a data set are arranged, especially when graphed. When looking at histograms, dot plots and stem-and-leaf plots, an overall pattern can be seen from the **shape of a distribution**.

The shape of a statistical distribution shows how the data is spread and can be seen by drawing a curve around the graph or display.

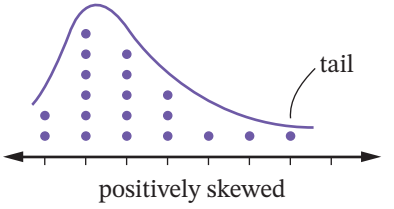
A distribution is **symmetrical** if the data is evenly spread or balanced about the centre.

For example, the frequency histogram and stem-and-leaf plot shown below are both symmetrical in shape.

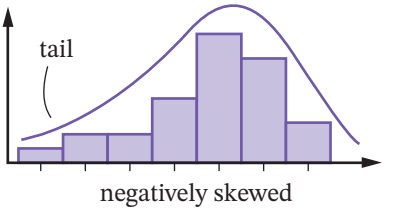


Stem	Leaf
2	6
3	3 7 9
4	5 8
5	0 2 3 3 7 8
6	0 4
7	6 7 8
8	9

A distribution is **skewed** if most of the data is bunched or clustered at one 'end' of the distribution and the other end has a 'tail', as shown on this dot plot and frequency histogram.



A distribution is **positively skewed** if its tail points to the right.

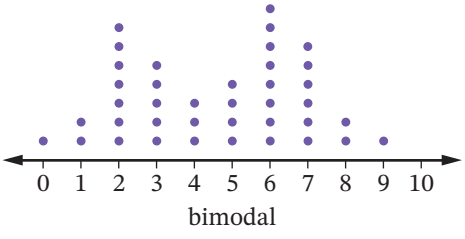


A distribution is **negatively skewed** if its tail points to the left.

A distribution is **unimodal** if it has one peak. All the above distributions are unimodal, and the peak indicates the mode, the data value with the highest frequency.

A distribution is **bimodal** if it has 2 peaks. The higher peak is the mode, while the other peak indicates another data value that has a high frequency.

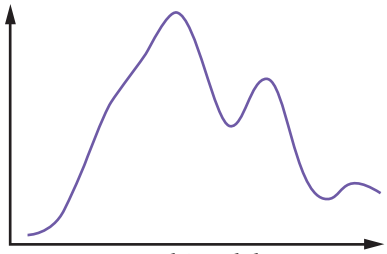
For example, this dot plot has peaks at 2 and 6 so it is bimodal. The mode, however, is 6.

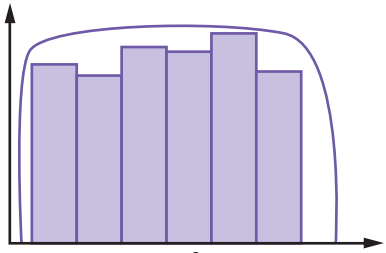


9780170479585

A distribution is **multimodal** if it has 2 or more peaks (including a bimodal distribution).

A distribution is **uniform** if all the data values have roughly the same frequency. There are no peaks, and its shape is rectangular.


multimodal


uniform

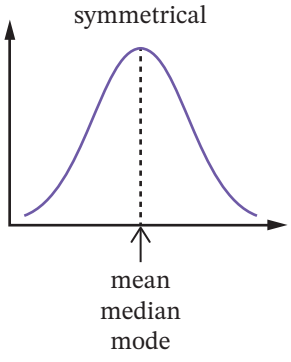
The **mode** of a statistical distribution is data value that has the highest peak of the distribution.

The **median** of a distribution is the data value that divides the area under the curve into half.

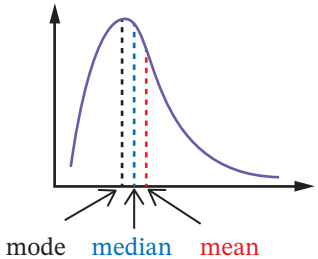
The **mean** of a distribution is most affected by outliers.

For a **symmetrical distribution**, the mean, median and mode are the same value.

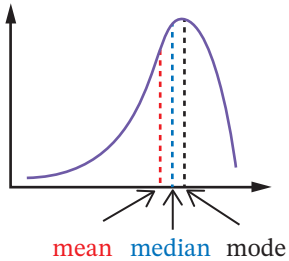
For a **skewed distribution**, the mean and median are pulled to the left or right so that the mean is further from the centre.



positively skewed



negatively skewed



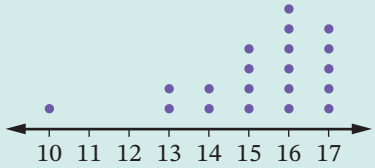
Example 8

For each statistical distribution:

i describe the shape

ii identify any outliers and clusters.

a



b

Stem	Leaf
3	0 1 2
4	1 3 4 4 5 6
5	0 4 5 7 8
6	3 7 8
7	0 1
8	4
9	8

Video
The shape of a frequency distribution

Solution

a **i** The shape is negatively skewed (tail points towards the lower values).

ii 10 is an outlier and clustering occurs at 15, 16 and 17.

b **i** The shape is positively skewed (tail points towards the higher values).

ii 84 and 98 are outliers and clustering occurs in the 40s and 50s.

EXERCISE 9.03 ANSWERS ON P. 637

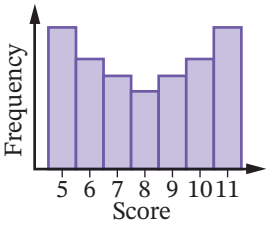
The shape of a distribution U F R C

EXAMPLE 8

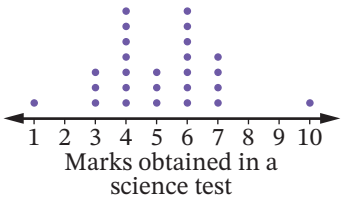
1 For each statistical distribution:

i describe the shape **ii** identify any outliers and clusters. C

a



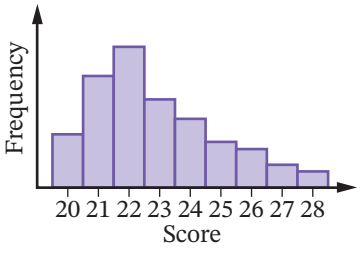
b



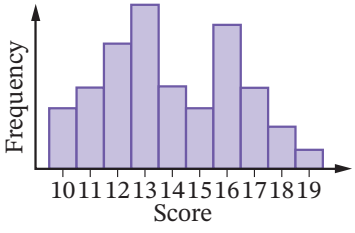
c

Stem	Leaf
2	0 2
3	1 4
4	2 5 6
5	2 3 6 7 8
6	0 4 4 9 9 9
7	1 2 4 5 5 6 7 8 8
8	2 4 5 6 7 7 8
9	0 2 2

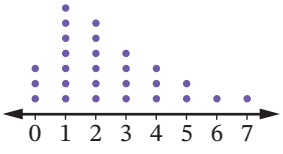
d



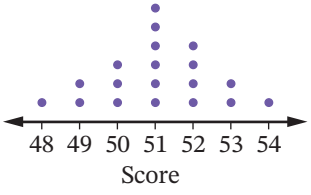
e



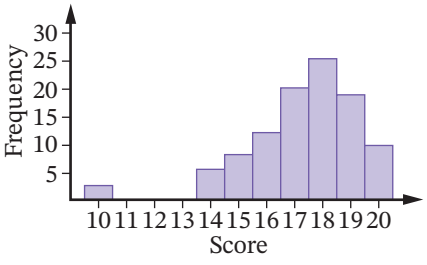
f



g



h



Foundation Standard Complex

9780170479585

2 The results of a Mathematics test for a Year 9 class are shown in the stem-and-leaf plot. R C

Stem	Leaf
1	8
2	3 5 8 8
3	2 4 6 7 8 8 9
4	0 4 5 5 7 7
5	3 6 7 8
6	0 4 4
7	2 5
8	1 2
9	2

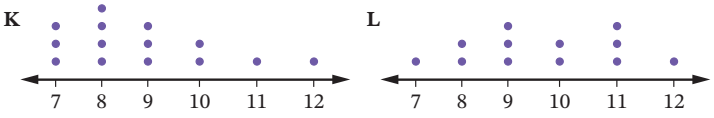
a How many students are in the class?
b Where does the clustering occur?
c Describe the shape of the data.
d Give a possible reason for the shape of the distribution.
e Find the mean, median and mode.

3 The number of children in the families of students at Eastleigh High School are shown below. C

4 4 7 3 3 6 8 6 3 3 4 2 9 5 1 2 3 3 4
7 2 7 2 2 3 5 3 1 4 1 4 1 2 5 6 3 5 2

a Arrange the data into a frequency table and construct a frequency histogram.
b Are there any outliers?
c Describe the shape of the distribution.
d Where does clustering occur?
e Find the mode, the mean (to one decimal place) and the median and show their position on the histogram.

4 Which statement is true about the 2 sets of data? Select the correct answer **A**, **B**, **C** or **D**.



A K is positively skewed and L is symmetrical.
B K is negatively skewed and L is bimodal.
C L has two peaks and K is positively skewed.
D K is positively skewed and the median of L is 9.

5 The results of an entrance exam to select apprentices for an energy supplier are shown in the stem-and-leaf plot. R C

Stem	Leaf
9	2 3 5 6 7 8
8	3 3 6 7
7	2 4 8
6	0 3 4 4
5	3 5 6 6 8 9
4	
3	
2	6

a Describe the shape of the distribution, excluding the outlier.
b What is the mode?
c Find the median.
d Find the mean, correct to one decimal place.
e Find the range.
f Is the range a good indicator of the spread of scores? Give reasons.

6 The average monthly maximum temperatures for Sydney and Melbourne are shown. R C

Month	J	F	M	A	M	J	J	A	S	O	N	D
Sydney	26	26	25	22	19	17	16	18	20	22	24	26
Melbourne	26	26	24	20	17	14	13	15	17	20	22	24

a **i** Display the data for Sydney and Melbourne on dot plots.
ii For each city, describe the shape of the dot plot.
b Find the mean (to 2 decimal places), median, mode and range for each city.
c Which city is generally warmer? Give reasons for your answer.

Foundation Standard Complex

9.04 Comparing data sets

Worksheets
Comparing word lengths
Investigating young drivers

Technologies
Population polygons
World climate

Interactive
World climate

Example 9

This back-to-back stem-and-leaf plot shows the results of a Year 9 class in a Health exam.

Girls		Boys
2	2	3 4 8
8 3 1	3	0 3 5 9 9
6 2	4	1 3 7 8 8 8 9 9
7 3 0	5	0 2 3 4 5 5 8 9 9
8 7 7 6 2 1	6	0 1 1 4 7 8
9 8 8 7 5 4 4 3	7	0 2 6 7
9 9 8 7 7 5 3 2 2 0	8	1 3 8
8 5 4 0	9	1 6

a Find the mean (to one decimal place) for:

i the girls **ii** the boys **iii** the whole group.

b Find the median for:

i the girls **ii** the boys

c Where are the scores clustered for:

i the girls? **ii** the boys?

d Compare the shapes of the 2 data sets.

e Which group, girls or boys, performed better? Justify your answer.

Solution

a i Mean for girls $= \frac{2600}{37}$

≈ 70.3

ii Mean $= \frac{2234}{40}$

≈ 55.9

iii Mean for group $= \frac{4834}{77}$

≈ 62.8

b i Median for girls $= 75$ 19th score

ii Median for boys $= \frac{54+55}{2}$ Average of the 20th and 21st scores

$= 54.5$

c i The girls' scores are clustered at 70s and 80s.

ii The boys' scores are clustered at 40s and 50s.

d The scores for the girls are negatively skewed, while the scores for the boys are positively skewed. The scores for the girls are generally higher than the scores for the boys.

e The girls performed better, which can be seen from their mean of 70.3 being greater than the mean of 55.8 for the boys. Also, the median for the girls was 75, which was higher than the median of 54.5 for the boys.

EXERCISE 9.04 ANSWERS ON P. 638

Comparing data sets U F R C

1 The times (in seconds) taken for swimming 100 metres by Brooke and Ryan are shown. R C

Brooke:	59.71	59.20	60.21	60.13	59.69	59.50	60.41	59.33	58.95	58.70
Ryan:	55.60	56.20	57.50	56.70	57.15	57.65	58.10	59.70	60.15	56.65

a Find the mean for each swimmer.
b Find the median for each swimmer.
c Find the range for each swimmer.
d Which swimmer is faster? Give reasons.
e Which swimmer is more consistent? Give reasons.

2 The maximum daily temperatures in Sydney and Adelaide for a 15-day period in March are shown. R C

Sydney	34.3	21.9	32.3	30.6	21.2	22.0	25.1	29.2
	31.5	29.2	25.9	30.6	32.1	27.3	24.7	
Adelaide	22.0	22.4	24.3	24.3	26.2	34.6	34.9	23.2
	22.1	21.6	26.8	32.0	28.3	27.1	29.0	

a Find the range for each city.
b Find, correct to 2 decimal places, the mean for each city.
c Find the median for each city.
d On how many days was Sydney warmer than Adelaide?
e Which city was warmer? Give reasons.

3 A class was given a topic test on trigonometry. The results were disappointing and, after some more work on the topic, the class was given a second test on trigonometry. The results of the 2 tests are shown in this back-to-back stem-and-leaf plot. R C

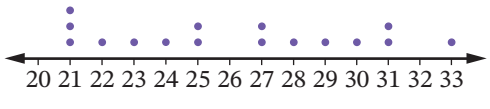
a Find the median for each test.
b Find, correct to one decimal place, the mean for each test.
c Find the range for each test.
d Describe the shape of each test.
e Has significant improvement been made from Test 1 to Test 2? Give reasons.

4 The dot plots show the temperatures (in °C) at 3 p.m. for Logan City, Queensland and Wollongong, NSW for 16 days in Autumn. R C

Logan City



Wollongong



9.04

EXAMPLE 9

Foundation | Standard | Complex

a Which city's temperatures are more skewed?

b Find, correct to 2 decimal places, the mean for each city.

c Find the range for each city.

d Which city had a more consistent temperature? Give reasons.

e Find the median for each city.

f Which city had cooler temperatures? Give reasons.

Test 1		Test 2
1	9	0 9
5 2	8	4 8 8
4 3 2	7	0 1 5 5 8 8
7 4 4 1 0	6	0 3 4 4 7 7 7 8
9 9 9 6 5 3 2 2	5	3 4 4 9 9
9 8 7 4 3 0	4	5 7 8
4 2	3	



AAP Image/DARREN ENGLAND

5 The reaction times (in seconds) of a group of men and women were measured. R C

Men:	38	29	37	28	30	32	31	53	31	26
	49	47	33	55	29	44	32	48	43	29
Women:	51	34	40	35	45	40	36	48	57	39
	42	35	51	54	36	37	37	39	43	45

a Arrange the 2 sets of data in a back-to-back stem-and-leaf plot.

b Would dot plots be suitable for displaying the 2 sets of data? Give reasons.

c Are the mean and median the same or different for:

i men? **ii** women?

d Find the mode for each group.

e Describe the shape of each distribution. Are there any outliers or clusters?

f Which group has the faster reaction times?

g Which group has the greater spread of reaction times?

6 Use the Internet to find the gold medal winning times for the 100 metre race for men and women for each Olympic Games from 1960 to 2021. R C

a Copy and complete this table by entering the winning times in seconds.

	1960	1964	1968	1972	1976	1980	1984	1988	1992	1996	2000	2004	2008	2012	2016	2021
Men																
Women																

Foundation Standard Complex

9780170479585

b Calculate the mean winning time (correct to 2 decimal places) for:
- **i** men
- **ii** women.

c Find the median for:
- **i** men
- **ii** women.

d Where are the data values mainly clustered for:
- **i** men?
- **ii** women?

e Identify any outliers for:
- **i** men
- **ii** women.

f Compare the shape of the 2 data sets.

g Did either group show a significant improvement in the winning time for the period 1960–2021? Identify any contributing factors.



Alamy Stock Photo/PCN Photography

9.04

Technology

Comparing relative humidities

Relative humidity is the amount of moisture in the atmosphere, measured as a percentage of water vapour per area at a certain temperature. It is reported more often in summer. When presented together with a high temperature, it shows that a day has been particularly warm and 'muggy'. In this activity, we will compare the relative humidity in Adelaide at 9 a.m. and again at 3 p.m. each day for the first fortnight in September in 2 consecutive years.

1 Enter the data into a spreadsheet (as shown).

	A	B	C	D	E	F	G
1	Sept 2018	Relative Humidity 9am Adelaide 2018 (%)	Relative Humidity 3pm Adelaide 2018 (%)		Sept 2019	Relative Humidity 9am Adelaide 2019 (%)	Relative Humidity 3pm Adelaide 2019 (%)
2	1	92	55		1	79	44
3	2	62	62		2	39	26
4	3	57	35		3	97	57
5	4	38	27		4	92	71
6	5	23	22		5	82	57
7	6	76	53		6	70	62
8	7	70	40		7	74	55
9	8	54	38		8	78	69
10	9	54	53		9	53	49
11	10	63	20		10	53	38
12	11	17	56		11	33	22
13	12	62	52		12	47	60
14	13	65	39		13	91	65
15	14	35	31		14	58	35

Foundation | Standard | Complex

2 Select **Scatter graph with smooth lines and markers** to graph this data on one set of axes. Give the graph a title and label the axes.

3 In B16, C16, F16 and G16, enter formulas to calculate the **average (mean)** relative humidity at 9 a.m. and 3 p.m. for each fortnight.

5 In B17, C17, F17 and G17, enter formulas to calculate the **maximum** relative humidity.

6 In B18, C18, F18 and G18, enter formulas to calculate the **minimum** relative humidity.

7 In B19, C19, F19 and G19, enter formulas to calculate the **range**.

8 In B20, C20, F20 and G20, enter formulas to calculate the **median**.

9 Compare your graphs as well as your statistical values. What do they suggest about the relative humidity each year? Is there a year and time of day showing a more consistent relative humidity? Justify your answer.

10 Visit the Bureau of Meteorology website **www.bom.gov.au** and select **Daily Weather Observations** to find the relative humidity for the past fortnight in your area. Compare the data with this time last year.

- **a** Can you see any patterns?
- **b** Is the relative humidity consistent?
- **c** Are there any outliers?
- **d** Analyse the data and describe any similarities or differences you can see.

Quiz
Mental skills 9

Mental skills 9: Maths without calculators ANSWERS ON P. 638

Finding a percentage of a multiple of 10

We can use the unitary method to calculate a percentage of a multiple of 10. To do this, we first find 1% of the number by dividing it by 100. This is easily done by moving the decimal point two places to the left.

1 Study each example.

a $1\% \times 300 = 3.00$
$= 3$

b $4\% \times \$500$
$1\% \times \$500 = \$5$
$\therefore 4\% \times \$500 = 4 \times \$5$
$= \$20$

c $9\% \times 40$
$1\% \times 40 = 0.4$
$\therefore 9\% \times 40 = 9 \times 0.4$
$= 3.6$

d $12\% \times \$700$
$1\% \times \$700 = \$7$
$\therefore 12\% \times \$700 = 12 \times \$7$
$= \$84$

2 Now evaluate each expression.

a	$1\% \times 800$	**b**	$1\% \times 30$	**c**	$3\% \times \$400$
d	$4\% \times 20$	**e**	$8\% \times \$600$	**f**	$3\% \times 40$
g	$7\% \times 80$	**h**	$2\% \times \$900$	**i**	$6\% \times \$50$
j	$8\% \times \$90$	**k**	$11\% \times 700$	**l**	$13\% \times 200$

3 Study each example.

a
$4\% \times 320$
$1\% \times 320 = 3.2$
$\therefore 4\% \times 320 = 4 \times 3.2$
$= 12.8$

b
$1.5\% \times \$200$
$1\% \times \$200 = \$2$
$\therefore 1.5\% \times \$200 = 1.5 \times \$2$
$= \$3$

c
$3\% \times 250$
$1\% \times 250 = 2.5$
$\therefore 3\% \times 250 = 3 \times 2.5$
$= 7.5$

d
$4.5\% \times \$800$
$1\% \times \$800 = \$8$
$\therefore 4.5\% \times \$800 = 4.5 \times \$8$
$= \$36$

4 Now evaluate each expression.

a	$5\% \times 250$	**b**	$1.5\% \times \$300$	**c**	$2\% \times \$540$
d	$9\% \times 110$	**e**	$2.5\% \times \$600$	**f**	$3\% \times 160$
g	$4\% \times \$750$	**h**	$4.5\% \times \$2000$	**i**	$3.5\% \times \$400$
j	$6\% \times \$120$	**k**	$3\% \times 190$	**l**	$1.5\% \times \$700$

Sampling and types of data 9.05

When the marketing department of a large company wishes to launch a new product, it collects information about consumer tastes. Governments collect information about the population so that they can provide for transport, housing, health and education needs. A school may need to collect information regarding school uniforms or food to be sold in the school canteen.

PATH
Puzzle
Data match-up
Videos
Counting crowds
Can fish oil make you smarter?
Can you trust your IQ?

Sample vs census

In statistics, the **population** refers to all the items under investigation or being studied. For example, if the life of AAA batteries is being tested, then the population is all the different brands or types of AAA batteries.

A **census** is a survey of the entire population. If the population is too large or difficult to survey, then a **sample** of the population is selected instead. The sample must be representative of the population, by being:

- **random:** every item in the population should have an equal chance of being selected for the sample
- **unbiased:** there should not be any factors that may influence the sample and prevent it from being a true indicator of the population
- **large:** the bigger the sample, the more representative it will be.

Example 10

Determine whether a census or a sample should be used for each investigation.

a Finding the number of primary school children in Victoria.

b Determining people's opinions on public transport.

c Finding the most streamed TV series in Australia.

PATH

Solution

a Census, because accurate figures are needed and these figures would be recorded.

b Sample, because it would be too expensive and time-consuming to survey all people.

c Census, because data can be collected about the number of accounts with a provider that streamed a certain TV series.

Types of data

There are 2 types of data: categorical and numerical.

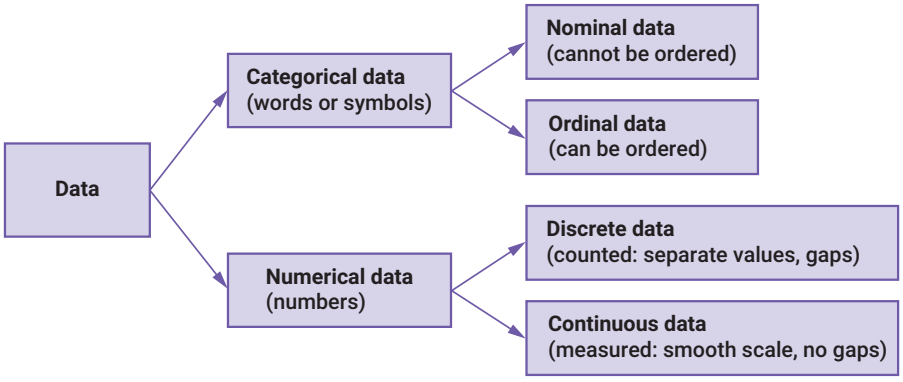
Categorical data are usually words and can be grouped into categories, such as a person's hair colour or cultural background. Categorical data is either nominal or ordinal.

- **Nominal data** cannot be ordered, for example, colour of car (nominal means 'name').
- **Ordinal data** can be ordered, for example, month of birth.

Numerical data (or quantitative data) are numbers describing things that can be counted or measured, like the number of goals scored or a person's height. Numerical data is either discrete or continuous.

- **Discrete data** are counted or measured and can only take on separate, distinct values, with 'gaps' or 'jumps', for example, the number of girls in a family is discrete because it could be 0, 1, 2, etc, but not 2.3 or $1\frac{2}{5}$.
- **Continuous data** are measured on a smooth scale with no 'gaps' or 'jumps' between values, e.g., temperature during the day is continuous because it can be 28°C, 29°C, 29.5°C, 30.1°, etc.

Worksheet
Surds on the number line



Example 11

Classify whether each type of data is categorical or numerical. If categorical, classify whether it is ordinal or nominal. If numerical, then classify whether it is discrete or continuous.

a School report grades (for example, A, B, C).

b The noise level at a concert.

c People's shoe sizes.

d Marital status (for example, single, married).

Solution

a	School report grades is ordinal categorical data.	Involves words or categories, can be ordered.
b	Noise level is continuous numerical data.	Takes on a smooth scale of values.
c	People's shoe sizes is discrete numerical data.	Takes on separate values.
d	Marital status is nominal categorical data.	Involves words or categories, cannot be ordered.

EXERCISE 9.05 ANSWERS ON P. 638

PATH

EXAMPLE 10

Sampling and types of data U F R C

1 Determine whether a census or a sample should be used for each investigation. R

- **a** Finding the number of people in Adelaide who were born overseas.
- **b** Testing the quality of running shoes.
- **c** Determining whether Australia needs a new flag.
- **d** Determining the number of people in Darwin who earn $60 000–$70 000 per year.
- **e** Determining the average family size in Australia.

2 In Australia, a national census was held in 2011, 2016 and 2021.

- **a** How often is a census held in Australia?
- **b** When will the next census be held?

3 The table shows the number of students in each year at Southwest High.

Year	7	8	9	10	11	12
Students	148	151	160	153	132	128

- **a** What is the school population?
- **b** What percentage (to one decimal place) of the school population is formed by:
  - **i** Year 8?
  - **ii** Year 11?
- **c** A sample of 100 students is to be surveyed on the amount of time that they spend on the Internet. How many students should be selected from:
  - **i** Year 8?
  - **ii** Year 11?

4 Students are surveyed about how they travel to school. The first 30 students arriving at school were selected for surveying. Is this a representative sample to use? Give reasons. R C

5 People were asked how often they eat take-away food. A sample of 100 people was surveyed at a shopping centre. Is this a representative sample to use? Give reasons. R C

Foundation Standard Complex

EXAMPLE 11

6 Classify each sample of data as either categorical or numerical. If categorical, state whether it is nominal or ordinal. R C

a The number of people absent from school today.

b The distance travelled by a car on a trip.

c The amount of petrol bought when filling up the car.

d The different types of pies available at the supermarket.

e The grade/belt colour of a Taekwondo student.

f The amount of time spent on devices during the week.

g The time for swimming 100 m.

h The country of birth of people living in Australia.

i The computer games played by students.

j The age you turn this year.

7 Students are asked to rate a movie on a scale of 1 to 5 stars. What type of data is this? Select **A**, **B**, **C** or **D**.

A continuous **B** categorical **C** discrete **D** random

8 Classify whether each type of numerical data is discrete or continuous. R C

a The number of people watching a cricket match.

b The temperature in Perth over a 24-hour period.

c The amount of time spent doing homework.

d The speed of a car passing through an intersection.

e The age you turn this year.

f The results in a maths test.

g The amount of rainfall in a month.

h The number of dogs at the RSPCA.

Did you know?

Australia's first census

The first census in Australia was conducted in November 1828. Before this, statistical information was collected by musters on the same day each year. Worried about the accuracy of the figures, Governor Darling and the Legislative Council passed an Act requiring a census to be held at regular intervals. Each citizen needed to disclose his or her name, age, marital status, number of children, the name of the ship on which they arrived, the amount of land owned and the number of cattle and other stock owned. Collectors were employed to gather this information by personal interview.

The first census showed that the total non-Aboriginal population was 36 598 and nearly 30% of that population lived in Sydney.

How many national censuses have there been in Australia?

Technology

Random numbers

Random numbers can be used to select people or items for a sample. A calculator or spreadsheet can be set up to generate a random number from 1 to 100, for example, and that number can be matched to the name of a person or item from a list. For a scientific calculator:

Casio	Sharp
Enter the formula RanInt#(1,100) as shown: ALPHA . 1 SHIFT) 100) =	2nd F 7 (RANDOM), select 3 R-INT, + 1 =

This generates a random number from 1 to 100. Press = more times to generate more random numbers.

On a **spreadsheet,** type **=INT(RAND()*100+1)** into a cell and press the ENTER key, then the F9 key repeatedly for more random numbers.

1. Use your calculator or spreadsheet to generate 12 random numbers from 1 to 100.
2. How many students are there in your class?
3. Use your calculator or spreadsheet to generate 12 random numbers from 1 to the number of students in your class.
4. Use a class roll to number the students in your class from 1 onwards. Suppose you need to choose a random sample of 10 students from your class for a survey on homework. Use your calculator or spreadsheet to generate 10 random numbers to select the 10 students for your survey.
5. Choose 4 students from each Year 9 maths class to answer some questions. Devise several different methods to ensure that your selection is random. Prepare a short class talk about your methods.

Investigation

Australian Bureau of Statistics

The Australian Bureau of Statistics (ABS) is the official organisation in charge of collecting data for government departments. The data collected covers many areas, from the height of children in Australia, the alcohol consumption in the country to the number of Internet subscribers and Internet providers. The ABS conducts a census of the entire Australian population every 5 years.

Visit the ABS website www.abs.gov.au to answer the following questions.

1. a What is a population clock?
   b What is the current predicted population of Australia?
   c On average, another baby is born in Australia after how many minutes?
   d On average, another person dies in Australia after how many minutes?
   e A new migrant arrives in Australia after how many minutes?
   f Australia's population increases by one after how many minutes?

2 Go to Australian Demographic Statistics and find the populations of the states and territories.
   a What percentage of Australia's population lives in:
      i NSW ii ACT?
   b Which state had the greatest percentage increase over the previous year?
3 Find the population pyramid for Australia's population.
   a What is a population pyramid?
   b What percentage of Australian males are aged 15–19?
   c What is the modal age?
   d How many people are aged 100 years and over?
   e Which group, males or females, have the higher life expectancy? Justify your answer.
4 a Explain why Australia's population is said to be ageing.
   b What is the median age of Australia's population?
   c Find the median ages of the people for all states and territories and display them in a table. Which has the:
      i highest median age? ii lowest median age?
5 Use the ABS website to find the number of people that live in your local area.

9.06 Bias and questionnaires

PATH

Video
The wrong guy won

Bias in sampling

When taking a sample, it is important that each item in the population has an equal chance of being chosen to be surveyed. This is called **random sampling**. To find a sample of students to survey about school uniform, we could place students' names in a box, mix them up, and then choose 100 names without looking. Alternatively, we could use a computer or calculator to generate random numbers to select students.

A sample that is not random is called a **biased sample**, and is not truly representative of the population. For example, if you survey the first 100 students to arrive at school today, or the first 100 students on the school roll, then not all students in the school have an equal chance of being selected. Your sample would not represent students who arrive at school later or whose names begin with letters at the end of the alphabet.

Example 12

A random sample of students is to be selected for their views on a new school uniform. Will each of the following sampling methods provide a sample that is representative or biased? If biased, explain why.

a Randomly selecting one student from each class
b Selecting 20 students from your mathematics class
c Selecting the students from the school's netball teams
d Randomly selecting students from the school roll by rolling a die and counting down the roll.

Solution

a	Representative.	Every student in the school has an equal chance.
b	Biased.	Because your mathematics class does not represent the other Year groups (or even the other Year 9 classes) at your school.
c	Biased.	Because the netball teams do not represent the whole school; it will also have mainly female students.
d	Representative.	Every student in the school has an equal chance.

Designing a questionnaire

Questionnaires are often used to collect information. They may ask for details about a person's opinions and attitudes, their lifestyle and behaviours, such as their recreational activities, TV viewing habits and types of foods eaten.

When designing a questionnaire, it is important that the questions asked are clear, fair and not biased.

- Decide what questions you want answered before you begin.
- Write the questions in plain English so that they are easily understood: make sure they are not vague, too open-ended or have a 'double meaning'.
- Ask for only one piece of information per question.
- Do not allow for too many different answers, provide choices if possible:
  - a tick or a cross beside each response
  - 'Yes', 'No', 'Don't know' or 'Other' responses
  - a rating on a scale
  - a one-word response
  - a sentence
- Determine how you are going to present and use the data you collect.
- Do not try to influence a particular answer unfairly (this is also called bias).
- Do not ask for information that may be too personal, such as phone numbers.

Example 13

Explain what is wrong with each survey question and suggest an alternative.

a 'How old are you?'

b 'How much pocket money do you receive each week?'

Solution

a The question is not precise enough. Possible answers could be:

15 15 years, 4 months 14 years 10 months and 3 days $16\frac{1}{2}$

A better question might be:

'What age do you turn this year?' or 'What is your date of birth?'

14 ☐ 15 ☐ 16 ☐ ...../...../..... (dd/mm/yy)

PATH

b The question is too general and responses could be:

Nothing $5/day $40/week Whatever I need

A better question could be:

'Circle the amount of pocket money you receive in a normal week:'

$5 or less $6 to $10 $11 to $15 more than $15

EXERCISE 9.06 ANSWERS ON P. 638

Bias and questionnaires U F PS R C

EXAMPLE 12

1 When collecting data, it is important to have an unbiased sample. Write, in your own words, what is meant by an 'unbiased sample'. C

2 List the possible areas of bias in each sampling method. R C

a A survey about the amount of time spent doing homework, taken at a football match.

b A survey about violence in sport, taken at a sporting event.

c A survey about the use of alternative fuels, taken at a car show.

d A survey about types of music listened to, taken at a rock concert.

3 A market research company uses a telephone survey to determine the most popular news program on TV. List some of the advantages and disadvantages of using this type of survey to collect data. C

4 In order to determine what type of music should be played on the school radio station, 20 students were given a questionnaire. PS R C

a Is the sample suitable for obtaining the required information? Give reasons.

b List some of the advantages of using a questionnaire to obtain the data.

c List some of the disadvantages of using this method.

d What other methods could be used to find out the type of music that should be played?

5 You need to determine how many students at your school have part-time jobs. Which of the following methods would be most suitable? Give reasons. C

A You ask 15 students in your class at school.

B You randomly select 15 students from each Year level at your school.

C You ask all students at your school to fill in a questionnaire and return them to you.

EXAMPLE 13

6 For each survey question, explain what is wrong with the question and suggest a better question. R C

a For how much time do you study?

b What is your favourite sport?

c What is your income?

d What do you have for lunch each day?

e What do you do in the holidays?

f How often do you exercise?

g How much time do you spend on social media websites?

h How many text messages do you send in one day?

Foundation Standard Complex

9780170479585

PATH

9.06

7 Select one topic for a questionnaire. PS R C

- Amount of TV watched at home.
- How students in Year 9 spend their time on a Sunday.
- The amount of time spent on homework during the week.
- The amount of time spent on the internet.
- Another topic of your choice.

a Design the questionnaire.
b Select a sample of students to answer it.
c Collect and present the information.
d Did the questions obtain the required data?
e Do you need to change any of the questions?
f Are there any questions you forgot to ask?
g Are there any questions you would omit next time?

8 Describe how you would collect data about each topic. PS C

a The most popular fast food eaten by students at your school.
b The average daily amount of money spent by students at your school canteen.
c The most popular TV series watched by students at your school.
d The most popular Australian artist listened to by students at your school.
e The average number of girls in a classroom in NSW.
f The most commonly used brand of toothpaste.

Investigation

Media reports of surveys

We live in a world of 24-hour news, often quoting the results of surveys. To detect bias in news reports, we need to always consider where the news item comes from and what types of samples the statistics are based on. Is the information supplied by a reporter, the police, a company executive, a government official or an opinionated blogger? Are the statistics based on a small sample, large sample, unrepresentative sample, or a phone or Internet poll of volunteers? Each may have a particular bias that may influence how the story is reported.

Find a report of a statistical survey from a newspaper, magazine or the Internet and investigate the following questions.

a Where did the article come from?
b Who wrote the article and what claims does it make about the population?
c Does it show any bias?
d What type of sampling was used? Was it representative?
e How many people were questioned?
f When was the survey conducted?
g Was there any bias in the sample?
h Was there any bias in the survey questions?

Foundation Standard Complex

Investigation

Year 9 student survey

The survey below is designed to collect data about Year 9 students. Questions can be changed or more can be added, but remember that the data is only useful if good questions are asked in a controlled way.

1 What is your gender? ☐ Male ☐ Female

2 What is your age (as of last birthday)? _____ years

3 How tall are you? _____ cm

4 How many people are in your household? _____ people

5 What is your position in your household (1 = oldest)? _____

6 How many pets do you have? _____ pets

7 How far from school do you live (to nearest 0.1 km)? _____ km

8 How did you travel to school today?
☐ Car ☐ Bus ☐ Walk ☐ Train

9 How long did the journey take? _____ minutes

10 On a scale of 0–10, rate your enjoyment of school (10 = most). _____

11 Which subject do you like best?
☐ Maths ☐ Science ☐ English
☐ Art ☐ PE ☐ History ☐ Other

12 Which subject do you study most?
☐ Maths ☐ Science ☐ English
☐ Art ☐ PE ☐ History ☐ Other

13 How much home study do you do each week? _____ hours

14 How many hours do you study maths each week? _____ hours

15 On a scale of 0–10, rate your enjoyment of maths (10 = most) _____

16 How many hours each week do you play sport out of school? _____ hours

17 How many sports do you play out of school? _____ sports

18 Which sport do you prefer to watch?
☐ Football ☐ Soccer ☐ Basketball
☐ Tennis ☐ Cricket ☐ Other

19 What is your favourite social network platform? _____

20 What is your favourite video streaming service? _____

Steps in conducting the survey

1 Finalise the questions and print the survey sheets.

2 Decide how many students you want to respond to the survey.

3 Decide how you will select the sample of students.

4 Decide whether the survey will be handed to selected students or if an interviewer will record responses.

5 Decide whether you will interpret the questions/responses or not.

6 Ask the selected students to complete the survey.

7 Collect all the sheets and label the first sheet 1, the second sheet 2 and so on.

8 Decide on the code you will use for each response, then code the responses and enter the results in a table, either on a spreadsheet or in your workbook.

9 Decide what to do if a question is not answered or the answer appears to be a mistake.

Power plus ANSWERS ON P. 639

1 The number of yoghurts sold daily in a school canteen were:

25	28	22	35	26	36	33	48	42	37	25	25	37	44	19	40	23
38	42	32	28	21	36	22	39	46	28	45	28	41	43	26	17	28

a Complete a frequency table by grouping the data into class intervals like this:

Class interval	Frequency
16–21	
22–27	
28–33	
34–39	
40–45	
46–51	

b Draw a frequency histogram and describe the shape, identifying any outliers or clusters.

c What is the modal class?

d Add a cumulative frequency column to your frequency table and find the median class.

2 When would it be impractical to use a stem-and-leaf plot?

3 Seeds were sown in 40 large boxes. The number of seedlings that sprouted per box were:

14	23	18	18	17	28	28	33	4	24
17	27	11	19	13	18	18	7	25	16
14	3	31	4	5	12	23	26	11	17
19	20	24	7	30	16	25	8	19	20

a Organise the data into a stem-and-leaf plot using intervals of:

i 10 for the stem **ii** 5 for the stem (0^0 for 0–4, 0^5 for 5–9, 1^0 for 10–14, ...)

b Describe the shape of both plots, identifying outliers and clusters.

c Which plot gives a better indication of how the seedlings have grown? Explain your answer.

CHAPTER 9 REVIEW

Quiz
Language of maths 9

Language of maths

bias	categorical data	census	cluster
continuous	cumulative frequency	discrete	distribution
histogram	mean	measure of centre	median
mode	negatively skewed	nominal	numerical data
ordinal	outlier	polygon	population
positively skewed	sample	stem-and-leaf plot	symmetrical

1 What is the collective name for the mean, median and mode?

2 Is a person's arm length an example of discrete data or continuous data?

3 What is the meaning of 'skewed'?

4 Explain the difference between a *population* and a *sample*.

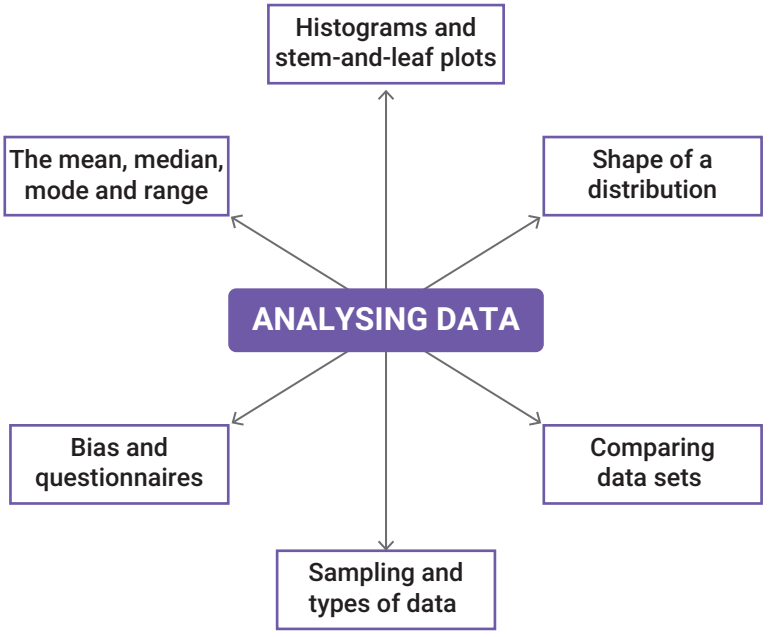
5 What name is given to an extreme value that is very different from the other values in a data set?

6 Name one way of avoiding bias when selecting a sample.

Topic summary

Quiz
Mind map: Analysing data

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



TEST YOURSELF 9

ANSWERS ON P. 639

Quiz
Test yourself 9

1 Find the mean, median, mode and range of this set of data. 9.01

8 6 9 4 10 8 5 5 8

2 A real estate agent sold 10 properties during one week. The prices of these properties are: 9.01

$420 000	$485 000	$379 000	$295 000	$1 750 000
$515 500	$315 500	$408 000	$368 000	$315 000

a Find the mean price of the properties.

b Find the median price of the properties.

c Compare the mean price with the median price. Which is the better measure of centre? Give reasons.

d When data on sales of properties are given, the median price is used, not the mean. Explain why.

3 These percentage marks were scored by 30 students in a Maths test. 9.02

45	67	78	54	22	38	58	64	73	89
84	70	78	62	56	59	42	47	86	90
64	63	72	77	43	85	75	71	70	83

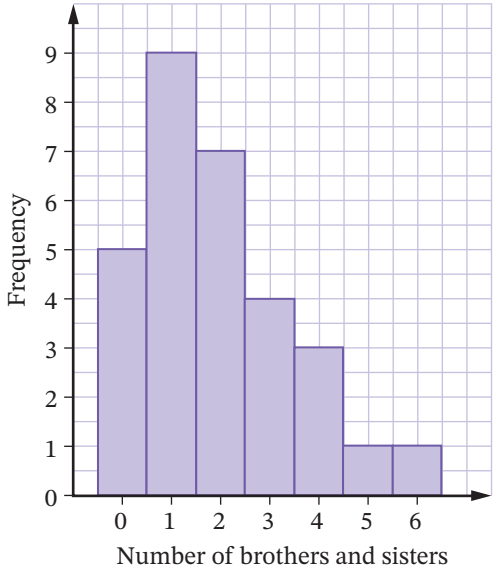
a Construct an ordered stem-and-leaf plot for the data.

b Find the mean, median, mode and range of the marks.

c Are there any outliers?

d Compare the mean and median. Which is more accurate? Give reasons.

4 A group of people were surveyed on the number of brothers and sisters they had. The results are displayed on the histogram. 9.02



a What is the range of this data?

b What is the mode?

c Find the mean, correct to 2 decimal places.

d What is the median?

e Which is the best measure of centre?

Foundation | Standard | Complex

9.02

5 24 patients were chosen at random from 2 medical centres. The waiting times (in minutes) for each patient are listed below.

Northbridge:	7	16	28	27	30	30	47	9	9	15	27	11
	38	35	42	53	23	22	45	32	37	37	18	16
Westvale:	44	13	7	9	25	26	37	35	8	18	18	15
	22	24	7	5	24	24	33	37	24	15	36	43

a Display the information in a back-to-back stem-and-leaf plot.

b Find the mean, median and range of both sets of data.

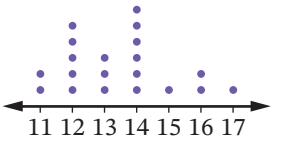
c Identify any outliers or clusters.

d Are there significant differences between the 2 sets of data? Explain your answer.

9.03

6 For each distribution, describe the shape and identify any outliers or clusters.

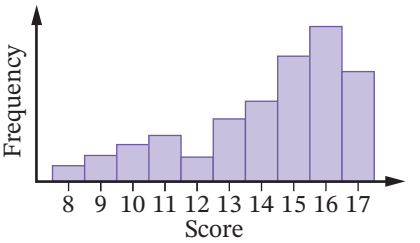
a



b

Stem	Leaf
8	7 8
9	1 3 4 4
10	0 2 7 8 8 9
11	1 2 2
12	0 7

c



9.04

7 The heights (in centimetres) of students in a Year 9 class are:

Girls:	160	153	155	156	142	163	162
	148	150	150	131	138	147	145
Boys:	150	165	164	178	170	171	164
	148	154	176	134	180	176	175

a Arrange the data in a back-to-back stem-and-leaf plot.

b Describe the shape of the display for:

i the girls **ii** the boys.

c Are there any outliers or clusters for either group?

d Find the mean (correct to 2 decimal places) and median for both groups.

e Which measure is the better indicator of the measure of centre?

f Which group is shorter?

g Which group has the greater spread of heights?

Foundation Standard Complex

9780170479585

PATH

9.05

8 Should a sample or a census be used to investigate each topic?

- **a** The most popular car colour
- **b** Whether energy-saving light globes really use less electricity
- **c** The number of retired people living on the Gold Coast, Queensland
- **d** People's views on which nation will win the soccer World Cup

9.05

9 Classify each type of data as categorical or numerical. If categorical, then classify it as nominal or ordinal. If numerical, then classify it as discrete or continuous.

- **a** The temperature in Alice Springs.
- **b** The number of views of a TikTok video.
- **c** Suburb or town of home.
- **d** The make of car passing through a motorway toll collection point.
- **e** The probability of rain today.

PATH

9.06

10 Explain why each of the following samples is biased.

- **a** A sample of online gamers were surveyed to investigate teenagers' views on road safety.
- **b** Customers in a café can give feedback about their meal by filling in a survey and posting it to the café.
- **c** At a factory, a sample of executives were surveyed about safety in the workplace.
- **d** A sample of audience members at a movie premiere were surveyed about their views on the film.
- **e** Classic Gold FM radio listeners were asked to vote for the best song of the year.

9.06

11 Explain what is wrong with each survey question and suggest an alternative.

- **a** Are you happy?
- **b** How did you enjoy your meal at our restaurant?
- **c** Australian soldiers have fought under our country's flag. Do you think that it should be changed?